

EXPLORATORY DATA ANALYSIS (EDA) REPORT

Project 2 – Data Analytics Internship

Submitted By

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1. Introduction

Exploratory Data Analysis (EDA) is the process of analyzing a dataset to understand its structure, identify patterns, detect outliers, and summarize important characteristics using descriptive statistics and visualizations. This project was completed using Microsoft Excel on the cleaned dataset prepared in Project 1.

2. Objective

The objectives of this project are:

- Calculate basic statistics such as count, mean, and median.
 - Analyze sales trends and customer purchasing behavior.
 - Identify outliers and unusual values.
 - Summarize key findings from the dataset.
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3. Dataset Description

The dataset contains customer order information collected from an e-commerce system.

Attribute	Description
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Total Records	1200
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Total Columns	14
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Dataset Type	E-commerce Sales Data
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The dataset includes the following fields:

- OrderID
 - Date
 - CustomerID
 - Product
 - Quantity
 - UnitPrice
 - ShippingAddress
 - PaymentMethod
 - OrderStatus
 - TrackingNumber
 - ItemsInCart
 - CouponCode
 - ReferralSource
 - TotalPrice
-

4. Descriptive Statistics

The following statistical measures were calculated using Microsoft Excel.

Metric	Formula Used
Total Orders	=COUNTA(OrderID)
Average Quantity	=AVERAGE(Quantity)
Median Quantity	=MEDIAN(Quantity)
Average Unit Price	=AVERAGE(UnitPrice)
Median Unit Price	=MEDIAN(UnitPrice)
Average Items in Cart	=AVERAGE(ItemsInCart)
Median Items in Cart	=MEDIAN(ItemsInCart)
Average Total Price	=AVERAGE(TotalPrice)
Median Total Price	=MEDIAN(TotalPrice)

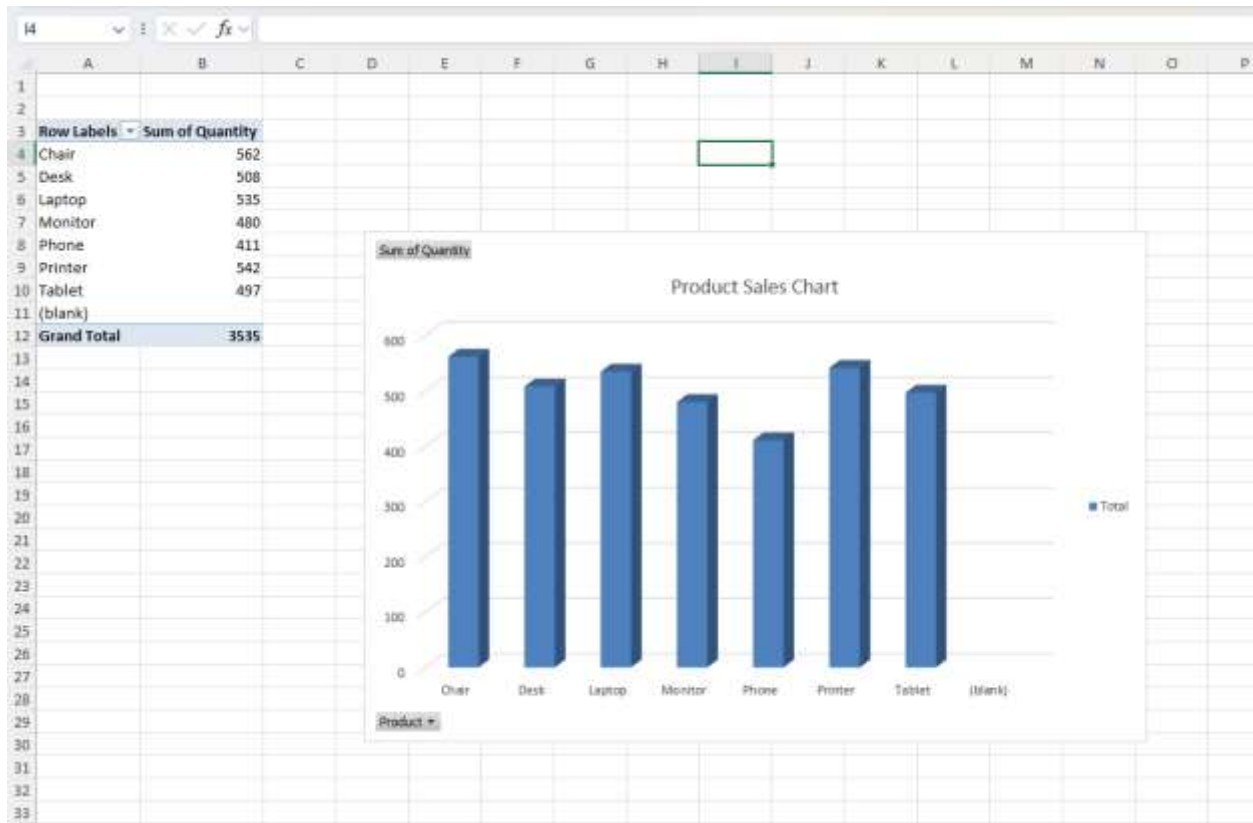
	A	B	C	D
1	Metric	Value		
2	Total Orders	1200		
3	Average Quantity	2.945833333		
4	Median Quantity	3		
5	Average UnitPrice	356.41275		
6	Median UnitPrice	364.21		
7	Average ItemsInCart	5.485		
8	Median ItemsInCart	5		
9	Average TotalPrice	1053.9683		
10	Median TotalPrice	823.615		
11				
12				
13				

5. Product Sales Analysis

A Pivot Table was created to analyze the quantity sold for each product.

Observation

The Pivot Table shows which products were sold most frequently and helps identify the best-selling products.

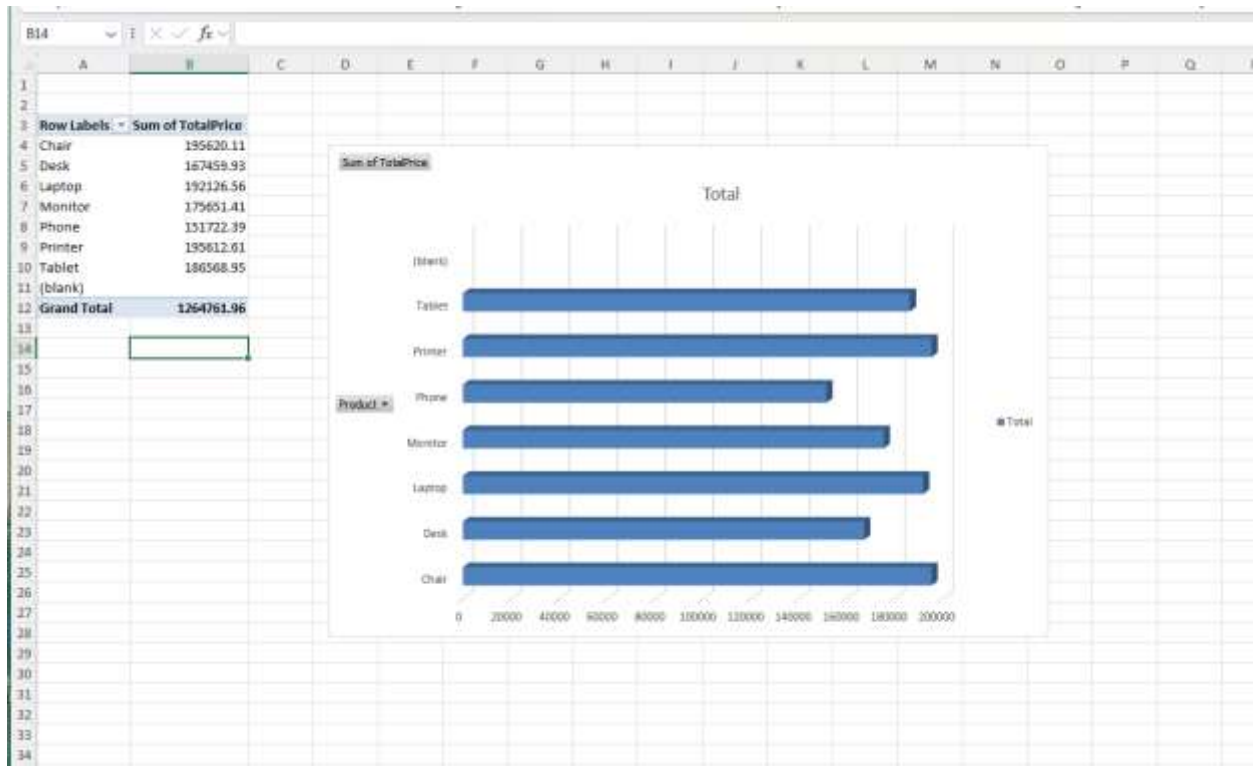


6. Revenue Analysis

Another Pivot Table was created to calculate the total revenue generated by each product using the **TotalPrice** column.

Observation

This analysis identifies the products that contributed the highest revenue.

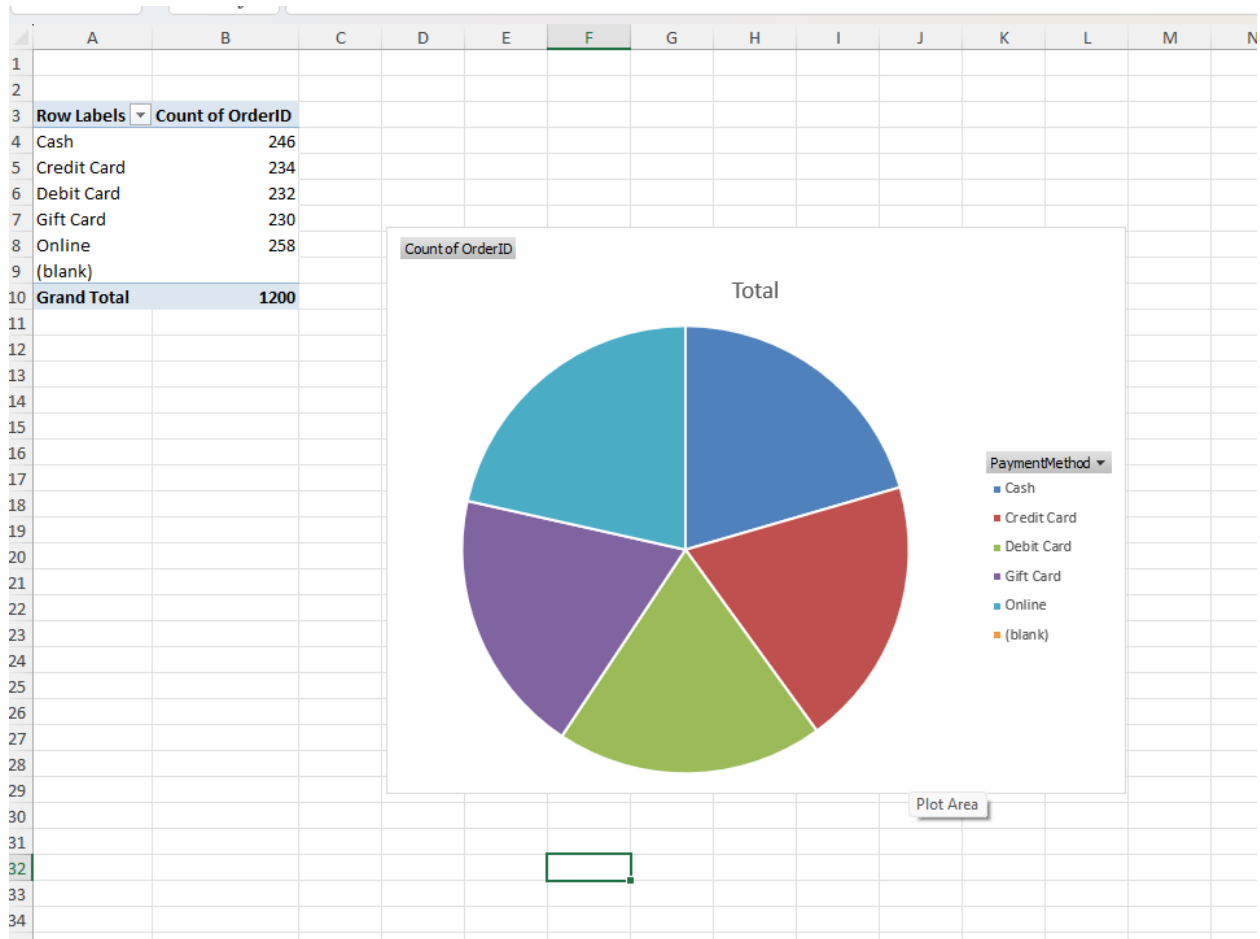


7. Payment Method Analysis

A Pivot Table and Pie Chart were created to analyze customer payment methods.

Observation

The analysis shows the distribution of payment methods such as Credit Card, Debit Card, Gift Card, and Online Payment.

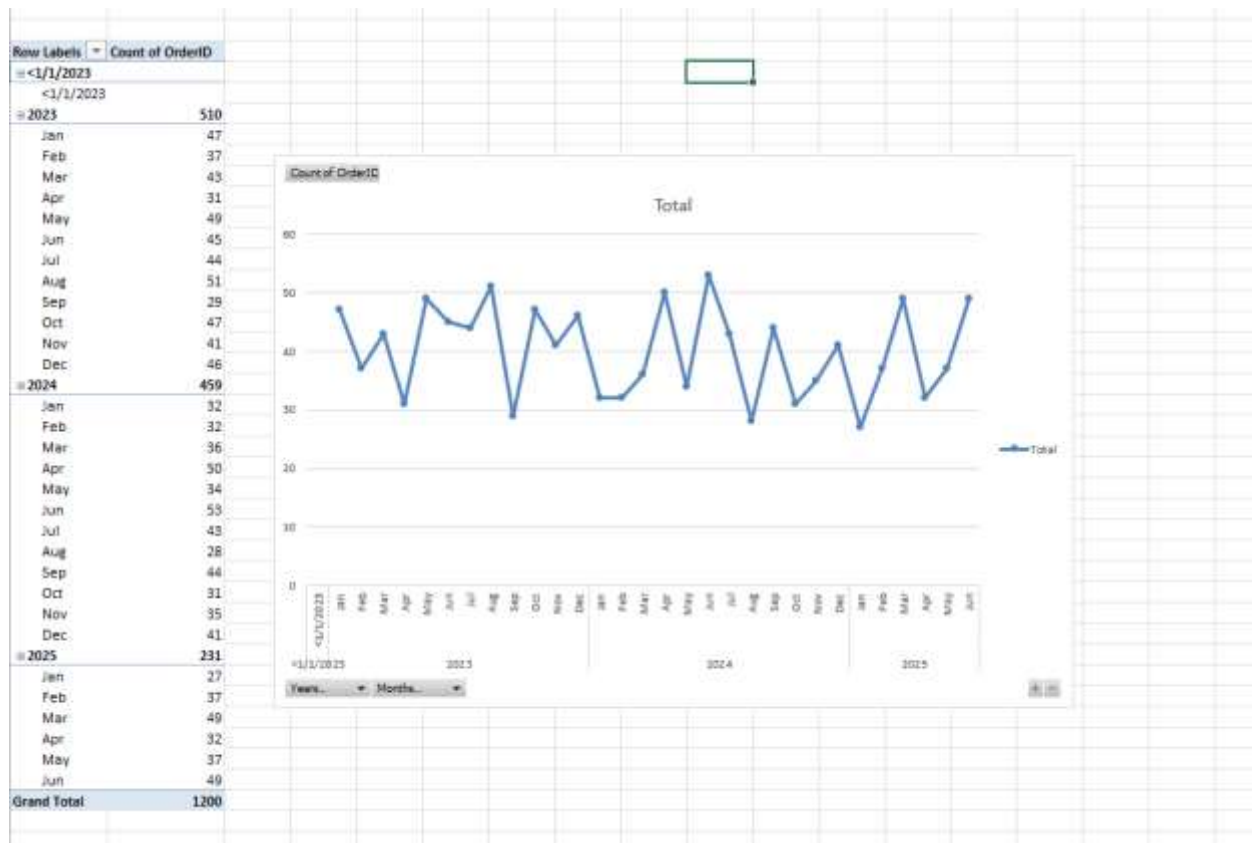


8. Monthly Sales Trend

The Date column was grouped by month using a Pivot Table, and a Line Chart was created.

Observation

The chart illustrates how the number of orders changes over time and helps identify months with higher or lower sales.



9. Outlier Detection

The **TotalPrice** column was analyzed using sorting and Conditional Formatting to identify unusually high-value transactions.

Observation

Several orders had significantly higher TotalPrice values compared to the majority of transactions. These records were considered potential outliers.

TotalPrice														
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1339	ORD000146	2025-03-04	C32182	Desk	2	661.89	551 Main St	Online	Cancelled	TR676287007	4	WINTER16	Referral	1339.67
1340	ORD001399	2024-03-06	C20099	Monitor	2	662.55	778 Main St	Online	Cancelled	TR688207889	5	No Coupon	Facebook	1340.06
1341	ORD000463	2023-05-26	C25276	Laptop	5	662.79	214 Main St	Debit Card	Shipped	TR14011732	3	SAVE30	Instagram	1341.9
1342	ORD000179	2025-04-04	C30218	Desk	4	662.94	535 Main St	Cash	Cancelled	TR686632861	8	SAVE30	Instagram	1342.76
1343	ORD000289	2023-11-29	C48942	Tablet	4	665.67	225 Main St	Credit Card	Pending	TR12555120	7	FREESHIP	Google	1343.88
1344	ORD000662	2023-10-06	C76675	Laptop	3	662.93	302 Main St	Gift Card	Returned	TR128042818	6	FREESHIP	Email	1344.79
1345	ORD000566	2025-08-17	C88891	Laptop	1	664.42	988 Main St	Gift Card	Returned	TR98417362	6	SAVE30	Facebook	1345.42
1346	ORD000173	2024-03-24	C39301	Phone	2	664.46	006 Main St	Online	Delivered	TR93839086	4	FREESHIP	Google	1346.26
1347	ORD000591	2023-02-26	C69182	Phone	6	664.51	114 Main St	Debit Card	Pending	TR116119704	8	SAVE30	Email	1347.55
1348	ORD000300	2023-02-17	C74966	Phone	1	664.79	745 Main St	Online	Delivered	TR93783265	8	No Coupon	Facebook	1348.37
1349	ORD000708	2025-03-04	C25266	Printer	1	665.11	786 Main St	Cash	Pending	TR125461216	3	FREESHIP	Instagram	1349.11
1350	ORD000894	2023-10-01	C90349	Printer	4	665.3	619 Main St	Cash	Shipped	TR132187596	9	FREESHIP	Email	1350.28
1351	ORD001065	2023-10-30	C47778	Printer	5	665.3	485 Main St	Debit Card	Delivered	TR179504223	7	SAVE30	Referral	1351.34
1352	ORD001136	2023-08-24	C71869	Tablet	2	666.23	799 Main St	Gift Card	Pending	TR124656005	4	WINTER16	Facebook	1352.99
1353	ORD000252	2024-08-20	C19640	Phone	4	666.58	621 Main St	Gift Card	Cancelled	TR931297879	5	FREESHIP	Facebook	1353.44
1354	ORD000525	2023-11-03	C54499	Monitor	1	666.48	160 Main St	Debit Card	Shipped	TR118061555	1	SAVE30	Email	1354.46
1355	ORD000979	2023-06-01	C76018	Monitor	6	668.04	298 Main St	Cash	Delivered	TR611794886	1	SAVE30	Facebook	1355.18
1356	ORD000528	2023-07-01	C68886	Laptop	5	670.48	273 Main St	Gift Card	Returned	TR196553867	5	SAVE30	Facebook	1356.4
1357	ORD000348	2023-07-02	C97187	Laptop	1	670.64	472 Main St	Cash	Shipped	TR162123713	6	FREESHIP	Referral	1357.64
1358	ORD000307	2023-09-27	C18776	Printer	9	670.79	848 Main St	Gift Card	Shipped	TR143892114	6	FREESHIP	Instagram	1358.79
1359	ORD000625	2023-09-24	C21123	Tablet	1	672.39	515 Main St	Credit Card	Pending	TR143250279	2	WINTER16	Email	1359.29
1360	ORD000338	2023-03-26	C18404	Tablet	6	674.04	546 Main St	Online	Cancelled	TR188401624	7	SAVE30	Google	1360.2
1361	ORD000201	2023-08-18	C10139	Monitor	2	674.9	148 Main St	Online	Shipped	TR178607531	4	SAVE30	Facebook	1361.8
1362	ORD000629	2024-04-07	C21181	Tablet	2	675.1	936 Main St	Cash	Pending	TR150369421	4	FREESHIP	Google	1362.2
1363	ORD000494	2023-09-02	C16917	Tablet	1	675.57	482 Main St	Cash	Shipped	TR161243924	1	WINTER16	Facebook	1363.57
1364	ORD000807	2024-11-10	C82344	Monitor	1	676.21	280 Main St	Online	Delivered	TR170957213	5	SAVE30	Google	1364.83
1365	ORD000469	2023-11-26	C12877	Chair	5	676.96	692 Main St	Cash	Cancelled	TR117254682	5	No Coupon	Facebook	1365.9
1366	ORD000885	2023-04-30	C72141	Monitor	4	677.37	158 Main St	Cash	Pending	TR176270481	8	SAVE30	Google	1366.48
1367	ORD000306	2023-04-02	C55464	Monitor	1	677.48	607 Main St	Online	Delivered	TR112161370	1	WINTER16	Facebook	1367.48
1368	ORD000889	2023-03-21	C99668	Monitor	1	678.01	686 Main St	Cash	Pending	TR167251261	4	FREESHIP	Referral	1368.01
1369	ORD000852	2023-09-02	C87100	Laptop	5	678.38	485 Main St	Gift Card	Delivered	TR138225104	7	WINTER16	Facebook	1369.8
1370	ORD001132	2023-06-07	C39040	Monitor	5	678.29	766 Main St	Online	Returned	TR132486970	6	No Coupon	Facebook	1370.06
1371	ORD000866	2025-02-14	C166786	Phone	1	678.88	607 Main St	Credit Card	Cancelled	TR174267180	4	SAVE30	Referral	1371.88
1372	ORD000789	2023-07-06	C13366	Printer	4	678.55	568 Main St	Cash	Delivered	TR190891430	5	No Coupon	Email	1372.4
1373	ORD000826	2024-03-26	C16409	Desk	1	679.66	390 Main St	Debit Card	Shipped	TR171888011	4	FREESHIP	Google	1373.66
1374	ORD000517	2024-06-25	C31494	Chair	2	680.85	953 Main St	Gift Card	Delivered	TR183312111	6	FREESHIP	Referral	1374.7
1375	ORD000719	2024-03-21	C46920	Laptop	2	681.06	703 Main St	Debit Card	Shipped	TR166403837	3	FREESHIP	Facebook	1375.12
1376	ORD000843	2023-11-18	C80004	Printer	8	684.54	161 Main St	Cash	Cancelled	TR164230718	8	WINTER16	Instagram	1376.43
1377	ORD001219	2024-11-24	C39024	Laptop	1	684.89	280 Main St	Credit Card	Delivered	TR124873568	1	FREESHIP	Email	1377.89
1378	ORD000421	2024-01-20	C94211	Printer	4	685.2	184 Main St	Gift Card	Cancelled	TR167206771	6	WINTER16	Google	1378.6
1379	ORD000576	2023-08-06	C99116	Laptop	4	686.41	178 Main St	Cash	Shipped	TR150711973	9	No Coupon	Referral	1379.41
1380	ORD001383	2025-06-16	C45104	Tablet	1	687.89	348 Main St	Cash	Returned	TR138559816	1	FREESHIP	Email	1380.89
1381	ORD000663	2023-02-05	C69886	Printer	3	690.3	747 Main St	Credit Card	Shipped	TR112902744	4	WINTER16	Google	1381.3
1382	ORD001358	2023-07-19	C20513	Laptop	4	690.79	580 Main St	Credit Card	Shipped	TR196254797	8	No Coupon	Referral	1382.12
1383	ORD000709	2023-08-17	C57176	Tablet	5	691.39	183 Main St	Online	Delivered	TR176889762	10	SAVE30	Email	1383.4
1384	ORD000771	2024-08-12	C20997	Phone	9	692.44	272 Main St	Debit Card	Cancelled	TR1882736458	6	FREESHIP	Google	1384.44
1385	ORD000935	2023-11-24	C39593	Phone	3	692.71	596 Main St	Credit Card	Delivered	TR162405251	3	FREESHIP	Instagram	1385.13

10. Key Findings

The analysis identified the following insights:

- The dataset contains **1200 customer orders**.
- Product sales vary across different product categories.
- Payment methods are distributed among multiple options.
- Monthly order volumes change over time.
- A few transactions have unusually high TotalPrice values, indicating possible bulk purchases or premium orders.
- The cleaned dataset is suitable for further analysis and dashboard creation.

11. Conclusion

This Exploratory Data Analysis (EDA) project successfully examined the cleaned e-commerce dataset using descriptive statistics, Pivot Tables, and charts in Microsoft Excel. Basic statistical measures such as count, mean, and median were calculated, while

trends, product performance, payment methods, and outliers were analyzed. The insights gained from this analysis improve the understanding of customer purchasing behavior and prepare the dataset for advanced analytics and business intelligence applications.

12. Tools Used

- Microsoft Excel
 - Pivot Tables
 - Pivot Charts
 - Excel Functions (COUNTA, AVERAGE, MEDIAN)
 - Conditional Formatting
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